8D Report - Oil Leakage at Gearbox Housing Flange (Customer Complaint)

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D1 Team

Team leader: quality engineer line L2. Members: process engineer assembly, maintenance technician dispensing cell, supplier quality engineer, production shift leader.

D2 Problem description

On 2026-04-08 the customer reported oil leakage at the flange between gearbox housing and motor housing on 7 e-drive units ED-210 found at their vehicle end-of-line test. All 7 units were built on line L2 between 2026-03-23 and 2026-03-29. Our own leak test (WI-PNT-005) had passed all 7 units.

D3 Containment actions

- 100 % retest of all units in stock and in transit (1,412 units): 9 additional units failed and were reworked.
- Customer stock sorted by our service team at the customer site: 3 more units found.
- Additional leak test with helium sniffing on line L2 until the root cause was fixed.

D4 Root cause

Occurrence: the sealant bead had interruptions caused by air in the dosing line after cartridge changes. Repeated E-205 alarms had been acknowledged without venting the pump. In addition, the M10 flange bolts tightened to 40 Nm did not give enough clamping force for the thin 2.5 mm bead after thermal cycling.

Non-detection: the leak test was started after only 20 minutes of curing. With partly cured sealant, small leakage paths were still closed and only opened after thermal cycling at the customer.

D5 Chosen corrective actions

1. Raise the M10 flange torque from 40 Nm to 45 Nm (engineering change ECN-2026-044).
2. Increase the sealant bead from 2.5 mm to 3.0 mm and reduce the open time from 15 to 10 minutes.
3. Add a camera-based bead inspection after dispensing.
4. Increase the curing time before the leak test from 20 to 30 minutes.
5. Venting procedure after every cartridge change added to TS-ROB-004.

D6 Implementation and verification

All actions were implemented on 2026-06-01 with WI-ASM-014 Rev C and WI-PNT-005 Rev D. Verification: 500 units with thermal cycling test (-40 to +120 degrees C, 50 cycles), zero leaks.

D7 Prevention

The lesson learned was shared with the plants in Brno and Pune. The dispensing FMEA was updated: occurrence for "bead interrupted" was reduced from 6 to 3 after the vision check.

D8 Closure

The customer accepted the 8D report on 2026-07-15. Cost of the complaint including sorting and rework: 86,400 EUR.

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